DSC6007 (Semester A, 2026)

Dynamic Programming and Reinforcement Learning

Group Project (3-4 persons per group)

1 Group Project: Components

You will work on a specific topic, which you need to do

- Short Presentation (2 mins) + Poster on **Nov 11 (week 10)** - Brief introduction to your project
  - Please prepare a poster for the presentation.
  - **Submit your poster on/before Nov 10, 6 pm**.
  - One speaker for this presentation.
  - NO READING SCRIPT PREPARED
- Presentation (15 mins)
  - Please prepare slides for the presentation.
  - You need to **record your presentation using ZOOM**. Do NOT edit the recording.
  - **Submit your slides and presentation on/before Nov 24, 6 pm**.
- Report
  - This should include everything you plan to say in the presentation.
  - Maximum 15 pages (You may put additional material in the Appendix if necessary).
  - **Submit your report on/before Nov 24, 6 pm**.
- Live Presentation (15 mins) + Q&A (5-10 mins) on **Nov 25 (week 12)**
  - *Few groups will be randomly draw for the live presentations.*
  - The content should be exactly the same as the 15 mins presentation.
  - Voting for the winners. There will be reward :)
  - Independent of your grade (except for participation).
  - NO READING SCRIPT PREPARED

- Code
  - Code that can reproduce your results.
- Demos (Optional)
  - Make sure you have the demos for the performance of before training, during training, and after training.
- Feedback
  - Ask questions to other presentations.
  - Provide anonymous feedback to other groups.
- What you should cover
  - Background, motivation, and the main idea
  - Theory and/or Application
  - Numerical experiments
  - Conculsion + limitation & strength of the methods + possible extensions (theory & algorithm & application)
- This group project is worth 30% of your grade (so total 30 points), in which
  - Presentation: 12 points
  - Content: 12 points
  - Participation: 6 points (*i.e.* feedback and questions)

2 Group Project: Submission and Timeline

Each group will be named "Group x", where $x \in \{1, 2, \ldots, 18\}$. Please submit the files using the file names that are provided below.

- **ON/BEFORE Oct 18**
  - Form your own group and submit on Canvas **(One submission per group is sufficient).**
  - **Anyone who does not have a group by that time will be assigned to a group.**
  - **File name: [name]_group.txt (*e.g.* PeterChan_group_.txt)**
- **ON/BEFORE Oct 25**

- Confirm topic on Canvas (discussion).
- Do not change your topic once you post on Canvas.

- **ON/BEFORE Nov 10, 6 pm**
  - Pdf file: poster,
    File name: Group_[group name]_poster.pdf (*e.g.* Group_1_poster.pdf)

- **ON/BEFORE Nov 24, 6 pm**
  - Pdf file: slides,
    File name: Group_[group name]_slides.pdf (*e.g.* Group_1_slides.pdf)
  - Pdf file: report,
    File name: Group_[group name]_report.pdf (*e.g.* Group_1_report.pdf)
  - MP4 files: any video/animation/demonstration that is used in the presentation
    File name: Group_[group name]_demo_[i]_[before/during/after].mp4
    (*e.g.* Group_1_demo_1_before.mp4, Group_1_demo_1_during.mp4, Group_1_demo_2_before.mp4,. . .)
  - Txt/py files (let me know if you need cpp, hpp, or others): code
    File name: Group_[group name]_c_[filename].txt (*e.g.* Group_1_c_main.txt, Group_1_c_solver.txt,. . .)
  - *** Zip file: Group_[group name].zip (*e.g.* Group_1.zip), but name your files as advised above.

3 Group Project: Topics [Each group should work on a unique topic]

We need to choose **either**

1. **Extension of course material:** including but not limited to the following
   - Partially observable Markov decision processes (POMDPs)
   - MDPs with continuous states, actions or time steps
   - Average Cost MDPs
   - Stochastic Shortest Path Problem
   - Deep Reinforcement Learning
   - Stochastic Approximation & RL methods
   - Inverse Reinforcement Learning

- (Advanced) multi-armed bandit problem

2. **Research topic:** Any **DP/RL** paper from 2015 to present, from
   - Conference on Neural Information Processing Systems (NeurIPS)
   - International Conference on Machine Learning (ICML)
   - Conference on Learning Theory (COLT)
   - International Conference on Learning Representations (ICLR)
   - International Conference on Artificial Intelligence and Statistics (AISTATS)
   - Journal of Machine Learning Research (JMLR)
   - IEEE Transactions on Automatic Control